GDP: Measuring Total Production and Income

ECON 101 – Principles of Macroeconomics

Muhebullah Karimzada

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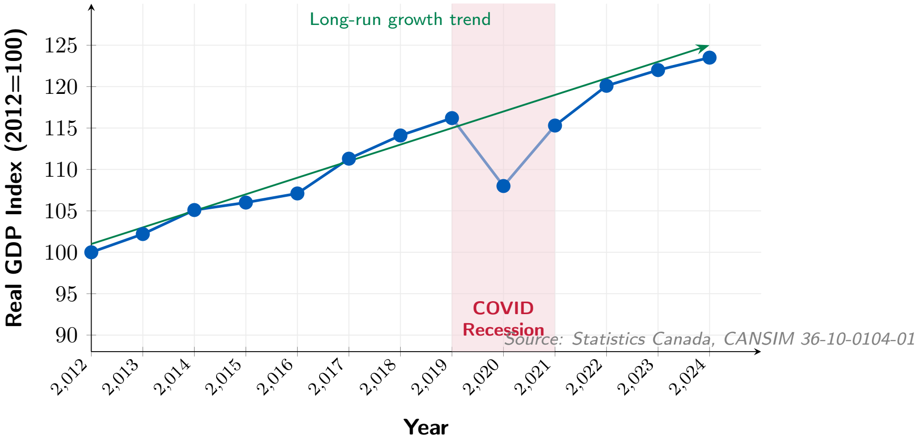
Chapter Outline

- **4.1** Gross Domestic Product Measures Total Production
- **4.2** Does GDP Measure What We Want It to Measure?
- **4.3** Real GDP versus Nominal GDP
- **4.4** Other Measures of Total Production and Total Income

Some Important Terms in Macroeconomics

- **Business cycle:** alternating periods of economic expansion and recession.
- **Expansion:** total production and employment are increasing.
- **Recession:** total production and employment are decreasing.
- **Economic growth:** the ability of the economy to produce increasing quantities of goods and services over the long run.
- **Inflation rate:** the percentage increase in the price level from one year to the next.

Figure 4.0 – Canada's Real GDP: Business Cycles and Long-Run Growth



Why These Concepts Matter for GDP

- GDP is our main measure of the **size of the economy**.
- Short-run business cycles show up as ups and downs in real GDP.
- Long-run economic growth shows up as an upward trend in real GDP.
- Inflation affects **nominal** GDP, so we must distinguish it from **real** GDP.
- **Canadian fact:** Canada's nominal GDP reached ≈\$3.0 trillion in 2024 (Statistics Canada, CANSIM 36-10-0104-01).

4.1 Learning Objective

Goal

Explain how total production is measured.

- **Gross domestic product (GDP):** the market value of all final goods and services produced in a country during a period of time.
- Unpacking the definition:
  - "**market value**" — uses prices as a common unit
  - "**final goods and services**" — avoids double-counting
  - "**produced in a country**" — location matters, not ownership
  - "**during a period of time**" — a flow, not a stock

“Market Value”

- We produce many different goods: lumber, canola oil, cars, haircuts, banking services. . .
- We cannot simply add *physical quantities* — a board-foot of lumber cannot be added to a barrel of oil.
- We need a common unit: **money**.
- We value each good at its market price, giving us the **market value** of production.
- **Canadian example:** In 2024, one barrel of Western Canadian Select oil ≈\$60 CAD; one tonne of potash ≈\$340 CAD — very different physical units, both measured in dollars.

Final Goods and Avoiding Double Counting

- A **final good or service** is purchased by a final user.
- **Intermediate goods and services** are inputs into other goods.
  - Example: wheat $\rightarrow$ flour $\rightarrow$ bread. GDP counts only the final bread sold to consumers.
- Counting intermediate goods would cause double counting.
- **Canadian example:** Alberta crude oil refined into gasoline sold at the pump — only the final gasoline sale counts; the crude oil is an intermediate good.

“Produced in a Country” and “During a Period of Time”

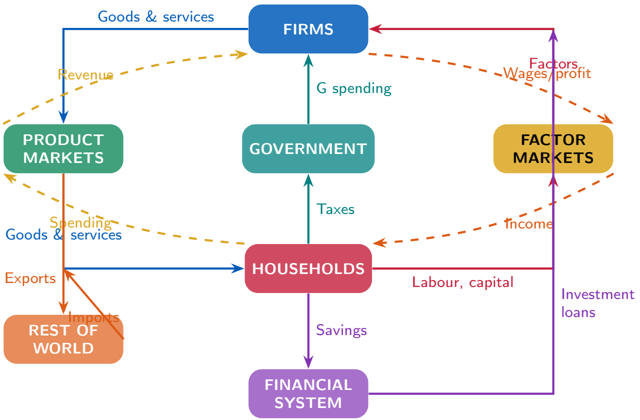
- GDP measures production within Canada’s borders, regardless of ownership:
  - A Toyota plant in Cambridge, ON $\Rightarrow$ **included** in Canadian GDP.
  - A Loblaws store in the Bahamas $\Rightarrow$ **excluded** from Canadian GDP.
- GDP is measured for a specific period (a year or quarter):
  - Only goods **produced** in that period count.
  - A used car re-sold in 2025 was already counted when new; it is **not counted again**.

Production = Income: Two Sides of the Same Coin

- Every dollar of production generates a dollar of income for someone:
  - workers receive **wages**,
  - capital owners receive **profit or interest**,
  - landowners receive **rent**.

$$\text{Total Production} = \text{Total Income}$$

- This identity underlies both the **expenditure approach** and the **income approach** to measuring GDP.


FIRMS
PRODUCT MARKETS
GOVERNMENT
FACTOR MARKETS
HOUSEHOLDS
REST OF WORLD
FINANCIAL SYSTEM
Goods & services
Revenue
G spending
Factors
Wages/profit
Taxes
Spending
Goods & services
Income
Exports
Imports
Labour, capital
Investment loans
Savings

The Four Sectors of the Circular Flow

- **Households & Firms**: the core of every market economy — households supply factors; firms produce goods.
- **Government (G)**: collects taxes; purchases goods; makes transfer payments (EI, OAS — not counted in GDP).
- **Rest of World**: exports (Canadian production sold abroad) and imports (foreign production bought in Canada). $\text{NX} = X - M$.
- **Financial System**: channels household savings to business investment and government borrowing (Bank of Canada, RBC, TD, etc.).

Income Approach to GDP

- Statistics Canada measures GDP by tracking **incomes earned in production**.

Component	**Share of GDP (2023)**
Compensation of employees	$\approx 51\%$
Wages and salaries	$\approx 44\%$
Employers' social contributions	$\approx 7\%$
Gross operating surplus	$\approx 27\%$
Net operating surplus	$\approx 14\%$
Capital consumption (depreciation)	$\approx 13\%$
Gross mixed income (small business)	$\approx 12\%$
Taxes less subsidies on production	$\approx 10\%$
GDP (income approach)	**100%**

Source: Statistics Canada, CANSIM 36-10-0104-01 (2023 estimates)

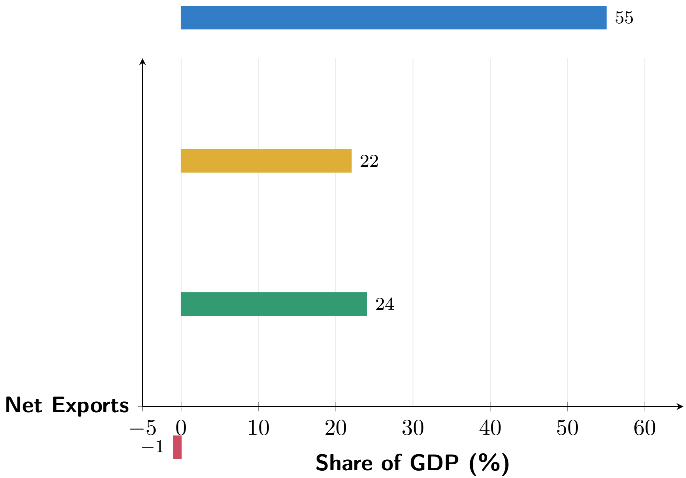
Expenditure Approach to GDP: $Y = C + I + G + NX$

- We can also measure GDP by tracking **spending** on final goods and services.

Component	**Symbol**	**Share of GDP (2023)**
Household final consumption	C	$\approx$ 55%
Government final consumption	G	$\approx$ 22%
Gross fixed capital formation (investment)	I	$\approx$ 24%
Net exports $(X - M)$	NX	$\approx$ -1%
Investment in inventories		$\approx$ 0%
GDP (expenditure approach)		**100%**

Source: Statistics Canada, CANSIM 36-10-0121-01 (2023)

Figure 4.2 – Components of Canadian GDP (2023)



Source: Statistics Canada, CANSIM 36-10-0121-01 (2023). Shares are approximate and rounded.

Value Added: An Alternative Way to Calculate GDP

- **Value added** at each stage of production = the firm's revenue minus the cost of intermediate inputs purchased from other firms.
- Summing value added across all firms gives total GDP.
- **Example (Canadian lumber to furniture):**
  - Timber harvester sells logs for \$200; value added = \$200.
  - Sawmill buys logs (\$200), sells lumber for \$350; value added = \$150.
  - Furniture maker buys lumber (\$350), sells chair for \$500; value added = \$150.
  - **Total value added = \$500 = final sale price.**

Goal

Discuss whether GDP is a good measure of well-being.

- GDP is a useful measure of total output.
- But it has **shortcomings** both as a measure of total production and as a measure of well-being.

What GDP Misses: Production Gaps

- **Household production**: childcare, cooking, home repairs — if done at home, not counted in GDP; if hired out, they are counted.
  - Statistics Canada's Household Satellite Account estimates household production was worth ≈$460B (41% of GDP) in 2015.
- **Informal (shadow) economy**: unreported legal activities and illegal transactions.
  - Estimated at ≈**13%** of official Canadian GDP — real value created but not captured.

What GDP Misses: Well-Being Gaps

- Even perfect GDP misses key dimensions of well-being:
  - **Leisure:** Canadians work $\approx$1,700 hours/year vs. the OECD average of 1,752 — leisure has real value.
  - **Pollution:** oilsands extraction raises GDP but degrades air and water quality; the cleanup cost is also GDP.
  - **Income distribution:** GDP per capita was $\approx$\$57,000 in 2023, but the Gini coefficient shows significant inequality.
  - **Crime and social problems:** more policing raises GDP even though the underlying problem is bad.

GDP and Well-Being: The Bottom Line

- GDP is an imperfect but **useful** indicator:
  - Higher GDP per person is strongly correlated with higher life expectancy, education, and happiness measures.
  - Canada's GDP per capita ≈$57,000 (2023) puts it among the top 20 worldwide (World Bank).
- Alternatives being developed:
  - **OECD Better Life Index** (Canada ranks #5 overall, 2024).
  - **Statistics Canada's Quality of Life Framework** (launched 2021): 84 indicators across 5 domains.
  - **UN Human Development Index:** combines income, education, and life expectancy.

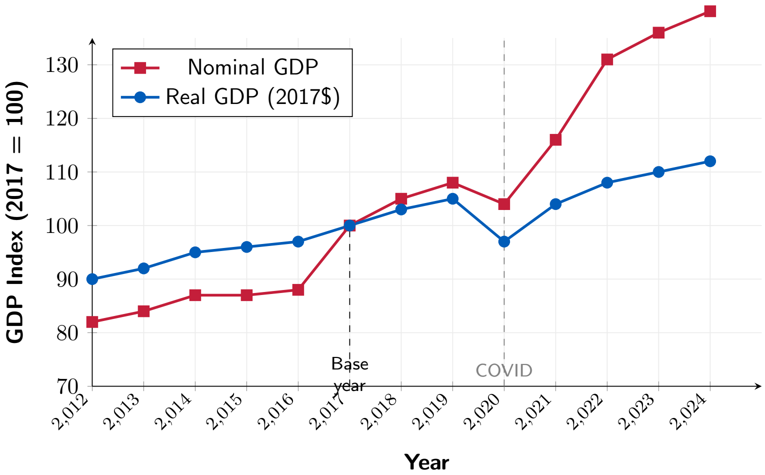
4.3 Learning Objective

Goal

Discuss the difference between real GDP and nominal GDP.

- GDP is measured in dollars — but dollar values change when prices change.
- We must separate **quantity changes** from **price changes**.
- **Nominal GDP:** values final output at current-year prices.
- **Real GDP:** values final output at base-year prices (Statistics Canada uses 2017 as reference year).

Figure 4.3 – Nominal vs. Real GDP: Canada 2012–2024



Source: Statistics Canada, CANSIM 36-10-0104-01. Approximate index values.

Calculating Real GDP: A Simple Example

- Suppose Canada produces only two goods: lumber and canola oil.

	Base Year (2017)		Current Year (2023)	
Good	Price	Qty	Price	Qty
Lumber (Mm^3)	$100	10	$130	12
Canola oil (Mt)	$500	5	$650	6
Nominal GDP	$3,500		$5,460	
Real GDP (2017$)	$3,500		$4,200	

- Nominal GDP overstates growth by counting higher prices.
- Real GDP isolates the increase in **actual quantities produced**.

Interpreting Real GDP

- In the base year, real and nominal GDP are **equal by definition**.
- After the base year:
  - If prices rise: nominal GDP grows faster than real GDP.
  - If prices fall: nominal GDP grows slower than real GDP.
- Growth rates reported in the news refer to **real GDP growth**.
- Statistics Canada uses **chain-weighted** prices (not a fixed basket) to avoid distortions as relative prices change over time.

The GDP Deflator

- The **GDP deflator** measures the overall price level of all final goods and services:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- In the base year: GDP Deflator = 100.
- **Canadian example:**
  - 2023: Nominal GDP ≈$2,890B; Real GDP (2017$) ≈$2,160B.

$$\text{Deflator}_{2023} = \frac{2{,}890}{2{,}160} \times 100 \approx 133.8$$

  - This means the overall price level rose ≈33.8% since 2017.

Other Measures of Total Production and Income

- Beyond GDP, national accounts construct:
  - **Gross National Income (GNI):** GDP plus net income earned by Canadians abroad minus income earned by foreigners in Canada.
  - **National Income:** GNI minus depreciation.
  - **Personal Income:** income received by households, including government transfers.
  - **Disposable Personal Income:** personal income minus taxes paid.
- **Canadian fact:** Canada's GNI is close to GDP since income flows in and out roughly balance.
- These measures are useful when studying **household budgets** and **living standards**.

Things to Remember

- GDP counts **final** goods and services only — to avoid double-counting intermediate inputs.
- GDP measures production **within borders** — not by whom the company is owned.
- **Total production = total income** — the circular-flow identity.
- **Real GDP** removes inflation; **nominal GDP** does not — always use real for quantity comparisons.
- GDP is a useful but imperfect proxy for well-being — it misses household production, leisure, inequality, and environmental quality.

Chapter Summary

- **GDP** = market value of all final goods and services produced in Canada during a period.
- Measured two ways (both yield the same answer): **income approach** and **expenditure approach** ($Y = C + I + G + NX$).
- GDP omits household production (≈41% of GDP) and the shadow economy (≈13% of GDP).
- **Real GDP** controls for inflation using base-year prices; Statistics Canada uses chain-weighting.
- The **GDP deflator** = (Nominal/Real) $\times$ 100 measures overall price level.
- Other measures: GNI, National Income, Personal Income, Disposable Income.

Thank you!

Questions?

Data: Statistics Canada CANSIM 36-10-0104-01, 36-10-0121-01; World Bank; OECD Better Life Index 2024